

Question	Answer	Mark	Mark scheme	Additional guidance
20	$x = 26.3\dots, y = -45.6\dots$ and $x = 1.71\dots, y = 3.57\dots$	M1	for correct substitution for y^2 or x^2 , eg $(7 - 2x)^2 = 3x^2 + 4$ OR for correct rearrangement and expansion of $(7 - 2x)^2$ to obtain 4 terms with all correct without considering signs or for 3 terms out of 4 correct with correct signs and substitution eg $(7 - 2x)^2 = 49 - 14x - 14x + 4x^2$ and $49 - 14x - 14x + 4x^2 = 3x^2 + 4$	NB $49 - 28x$ or $-28x + 4x^2$ can be considered 3 terms out of 4 correct with correct signs
		M1	for method to write a correct simplified equation eg $x^2 - 28x + 45 (=0)$	The quadratic does not have to equal 0, ie accept $x^2 - 28x = -45$
		M1	for a method to solve a correct quadratic eg $\frac{28 \pm \sqrt{(-28)^2 - 4 \times 1 \times 45}}{2 \times 1}$ or $\frac{28 \pm \sqrt{604}}{2}$ or $14 \pm \sqrt{151}$ or $(x - 14)^2 - 14^2 + 45 = 0$ oe	Can be implied by both x values correct or both y values correct
		A1	$x = 26.2$ to 26.3 , $y = -45.6$ to -45.5 and $x = 1.7$ to 1.712 , $y = 3.5$ to 3.6	Answers must be correctly paired (Maybe in the body of the working) If answers are given in the range in working and then rounded incorrectly award full marks